SE-2317 ENTRY BLACKOUT STATE ESTIMATION

Gao Jiang

DOC NO:	TR-2025-001
DATE:	2025-12-28
REV:	A
STATUS:	DRAFT

ABSTRACT

This report details the design and validation of a Tightly-Coupled Square-Root Error-State Kalman Filter (SR-EKF) for the SE-2317 entry vehicle. The architecture features a fixed-lag history buffer with rewind–update–replay semantics to handle delayed and out-of-order sensor data, alongside a robustness layer employing χ^2 gating and heavy-tailed variance inflation. To mitigate observability loss during the critical GNSS blackout phase, a dedicated pre-blackout accelerometer bias calibration strategy is implemented. Monte Carlo analysis (N=500) demonstrates a **100.0% pass rate** and consistent NEES/NIS performance, validating the system's resilience against sensor dropouts, outliers, and prolonged signal denial.

PROPRIETARY INFORMATION: The information contained in this document is proprietary to the organization and shall not be disclosed to any third party without prior written consent.

DOCUMENT CONTROL

Program	SE-2317 — Entry Blackout Estimation
Doc Type	Technical Report
Revision	A (Draft)
Owner	Gao Jiang

Rev	Date	Author	Change Summary
A	2025-12-28	Gao Jiang	Initial release with Monte Carlo evidence and traceability.

EXECUTIVE SUMMARY

Objective. Estimate 6-DOF entry state through a GNSS blackout window and meet the post-blackout (2 s) accuracy and consistency targets in `tasks.md`.
Deliverable. A research-grade but flight-structured estimator with traceable math, stable numerics, and quantified Monte Carlo performance.

- **Estimator.** A square-root error-state EKF (SR-EKF) on a 24D augmented state: $(r, v, q, \omega, b_g, b_a, f_{\text{res}}, \tau_{\text{res}})$, with multiplicative quaternion error injection.
- **Delayed / out-of-order sensors.** A fixed-lag history buffer with rewind–update–replay semantics. For linear-Gaussian models this is equivalent to stochastic cloning/factor-graph smoothing on a fixed window; for EKF it is first-order equivalent and empirically stable.
- **Robustness.** Innovation gating via χ^2 tests (NIS), plus a tight-coupling pseudorange down-weighting rule equivalent to an M-estimator (Huber-like) implemented as variance inflation.
- **Covariance realism.** Phase-dependent process-noise inflation and blackout-specific aerodynamic residual excitation to prevent overconfidence where model error dominates.
- **Bias strategy.** A dedicated pre-blackout accelerometer-bias calibration stage (batch normal equations accumulated online) applied at blackout entry to mitigate loss of bias observability during blackout.
- **Monte Carlo evidence (500 runs).**
  - **Tight-coupled GNSS:** pass rate **100.0%** (500/500) under 5% heavy-tail outliers and 20% dropout; mean NEES ≈ 4.89 with ≈ 0.948 coverage.
  - **PVT baseline:** pass rate **97.0%** (485/500); rare catastrophic inconsistency events dominate mean NEES.

1 PROBLEM SETTING AND REQUIREMENTS

The mission context is a 6-DOF rigid-body entry with strong aero nonlinearity and a GNSS blackout. Sensors are asynchronous with fixed delays (star tracker, radar altimeter) and stochastic dropouts and outliers. The primary acceptance window is the **2 seconds after blackout exit** ($t \in [t_{\text{exit}}, t_{\text{exit}} + 2]$, where $t_{\text{exit}} = t_0 + 160$ s).

Key quantitative targets (summary). Within the post-blackout window: position $3\sigma \le 40$ m, velocity $3\sigma \le 0.5$ m/s, attitude $3\sigma \le 0.15^\circ$, bias bounds $\|b_g\|_{3\sigma} \le 0.005^\circ/\text{s}$ and $\|b_a\|_{3\sigma} \le 120$ μg, and consistency via NEES/NIS 95% coverage ≥ 90% over Monte Carlo.

2 STATE, DYNAMICS, AND NOISE MODELING

2.1 State and error-state parameterization

We estimate the augmented state

$$x := \left[r,\ v,\ q,\ \omega,\ b_g,\ b_a,\ f_{\text{res}},\ \tau_{\text{res}}\right], \qquad r, v, \omega, b_g, b_a, f_{\text{res}}, \tau_{\text{res}} \in \mathbb{R}^3,\ q \in \mathbb{H},\ \|q\| = 1. \tag{1}$$

The EKF operates on an error state $\delta x \in \mathbb{R}^{24}$ with attitude error represented by a 3-vector $\delta\theta$ such that the corrected quaternion uses a multiplicative injection:

$$q^{+} = \exp(\tfrac{1}{2}\,\delta\theta) \otimes q^{-}, \qquad \exp(\tfrac{1}{2}\,\delta\theta) \approx \left[\cos(\tfrac{\|\delta\theta\|}{2}),\ \hat{\delta\theta}\sin(\tfrac{\|\delta\theta\|}{2})\right]. \tag{2}$$

This choice avoids quaternion normalization drift and yields a locally Euclidean error space for covariance.

2.2 Discrete-time propagation model

Let IMU measurements at time step k be $(\omega_{m,k}, a_{m,k})$. Bias-compensated signals are

$$\omega_k = \omega_{m,k} - b_{g,k}, \qquad a_k = a_{m,k} - b_{a,k}. \tag{3}$$

We propagate attitude by integrating ω_k over Δt and normalize:

$$q_{k+1} = \text{normalize}\Big(q_k \otimes \exp(\tfrac{1}{2}\,\omega_k \Delta t)\Big). \tag{4}$$

Define the body-to-inertial rotation $C(q) \in \text{SO}(3)$, and inertial gravity $g(r)$. The translational update is second-order in Δt:

$$v_{k+1} = v_k + \big(C(q_k)\,a_k + g(r_k) + a_{\text{learn}}(k)\big)\,\Delta t, \tag{5}$$

$$r_{k+1} = r_k + v_k \Delta t + \tfrac{1}{2}\big(C(q_k)\,a_k + g(r_k) + a_{\text{learn}}(k)\big)\,\Delta t^2. \tag{6}$$

Here a_{learn} is an optional bounded online correction (see §**??**).

2.3 Process noise: IMU, bias RW, and aero residual OU

We model gyro/accel measurement noise as white with spectral densities σ_ω and σ_a, and biases as random walks:

$$b_{g,k+1} = b_{g,k} + w_{bg,k}, \quad b_{a,k+1} = b_{a,k} + w_{ba,k}, \quad w_{bg,k} \sim N(0, \sigma_{bg}^2 \Delta t\, I),\ w_{ba,k} \sim N(0, \sigma_{ba}^2 \Delta t\, I). \tag{7}$$

Aero residuals. To model dominant entry mismatch, we include residual force/torque states with Ornstein–Uhlenbeck (OU) discretization:

$$f_{\text{res},k+1} = \phi_f f_{\text{res},k} + \eta_{f,k}, \qquad \tau_{\text{res},k+1} = \phi_\tau \tau_{\text{res},k} + \eta_{\tau,k}, \quad \phi_f = e^{-\Delta t/\tau_f},\ \phi_\tau = e^{-\Delta t/\tau_\tau}, \tag{8}$$

with $\eta_{f,k} \sim N(0, \sigma_f^2(1 - \phi_f^2)I)$ and $\eta_{\tau,k} \sim N(0, \sigma_\tau^2(1 - \phi_\tau^2)I)$. This yields a stationary marginal variance $\sigma_f^2 I$ and $\sigma_\tau^2 I$.

Discrete translational acceleration noise. Assume a white specific-force disturbance $w_a \sim N(0, \sigma_{a,\text{eff}}^2 I)$ affecting $\dot{v}$. Then the standard continuous-to-discrete integration gives the (RV) block:

$$Q_{\{r,v\}}(\Delta t) = \sigma_{a,\text{eff}}^2 \begin{bmatrix} \frac{\Delta t^3}{3} I & \frac{\Delta t^2}{2} I \\ \frac{\Delta t^2}{2} I & \Delta t\, I \end{bmatrix}. \tag{9}$$

We use $\sigma_{a,\text{eff}}^2$ (IMU noise plus model residual) and apply a phase-dependent scaling (pre/blackout/post) to enforce covariance realism where the model is weakest. We define the effective acceleration variance explicitly:

$$\sigma_{a,\text{eff}}^2 := \sigma_a^2 + \sigma_{\text{res}}^2. \tag{10}$$

3 OBSERVABILITY ANALYSIS AND BIAS MANAGEMENT

3.1 Local LTV observability (definition)

Let the linearized, time-varying error dynamics be

$$\delta x_{k+1} = F_k\, \delta x_k, \qquad y_k = H_k\, \delta x_k. \tag{11}$$

Define the state transition $\Phi_{k,0} = F_{k-1} \cdots F_0$. The system is observable on $k \in [0, N]$ if the stacked matrix $O_N := \begin{bmatrix} H_0 \\ H_1 \Phi_{1,0} \\ \cdots \\ H_N \Phi_{N,0} \end{bmatrix}$ has full column rank. Equivalently, for weighted measurements with covariance R_k, the discrete observability Gramian

$$W_N := \sum_{k=0}^{N} \Phi_{k,0}^{\mathsf{T}} H_k^{\mathsf{T}} R_k^{-1} H_k \Phi_{k,0} \tag{12}$$

is symmetric positive definite. We use this definition to avoid over-claiming a closed-form condition in a strongly coupled entry model.

3.2 Unobservable subspace: Accelerometer bias and attitude coupling

The translational error-state dynamics contain both accelerometer bias and attitude error terms:

$$\delta v_{k+1} \approx \delta v_k - C_k\, \delta b_{a,k}\, \Delta t - \left[C_k a_k\right]_\times \delta\theta_k\, \Delta t + \cdots. \tag{13}$$

To show why bias is <u>not</u> generically identifiable without excitation, consider a simplified continuous-time, constant-excitation model with constant C and constant specific force f^n (in navigation frame), and ignore gyro-error dynamics:

$$\delta\dot{v} = -[f^n]_\times\, \delta\theta - C\, \delta b_a, \qquad \delta\dot{\theta} = 0, \qquad \delta\dot{b}_a = 0, \qquad y = \delta v. \tag{14}$$

Lemma. The above system is not observable; in particular, it has a non-trivial unobservable subspace.

Proof. Let $\delta\theta \neq 0$ be arbitrary and set $\delta b_a = -C^{-1} [f^n]_\times\, \delta\theta$ with initial $\delta v(0) = 0$. Then $\delta\dot{v}(t) = 0$, hence $y(t) = \delta v(t) \equiv 0$ for all t despite $\delta\theta \neq 0$ and $\delta b_a \neq 0$. Therefore the system cannot be observable. *QED*

Implication. Separating accelerometer bias from attitude error requires <u>time variation</u> (in C_k and/or in the direction of f) so that information accumulates in the LTV Gramian W_N. Entry dynamics provide such excitation, but only sensors that constrain δv (GNSS, tight-coupled Doppler) can turn that excitation into bias information.

3.3 Blackout implication and design response

During blackout, GNSS-derived constraints are absent; the bias sub-system loses direct observability, and any bias correction must come from (i) pre-blackout information, (ii) post-blackout reacquisition, and (iii) conservative covariance growth. Therefore we adopt two explicit design decisions:

1. **Pre-blackout bias calibration** to reduce bias uncertainty before observability is lost.
2. **Blackout covariance realism** via increased Q (and aero residual OU excitation) to prevent "too-certain" predictions.

3.4 Pre-blackout accelerometer-bias calibration as MAP least squares

We treat a short pre-blackout interval as a dedicated calibration window. Let (r_k, v_k) be the estimator INS solution and $(\bar{r}_k, \bar{v}_k)$ the GNSS RV measurement at time t_k with covariance R_k. Linearize the RV residual with respect to a constant accelerometer-bias perturbation δb_a and initial RV offsets:

$$\begin{bmatrix} \bar{r}_k - r_k \\ \bar{v}_k - v_k \end{bmatrix} \approx \underbrace{\begin{bmatrix} I & (t_k - t_0)I & S_{r,k} \\ 0 & I & S_{v,k} \end{bmatrix}}_{A_k} \underbrace{\begin{bmatrix} \delta r_0 \\ \delta v_0 \\ \delta b_a \end{bmatrix}}_{\theta} + \varepsilon_k, \qquad \varepsilon_k \sim N(0, R_k), \tag{15}$$

where $(S_{r,k}, S_{v,k})$ are the accumulated RV sensitivities to δb_a along the propagated trajectory. We compute a MAP estimate with diagonal Gaussian priors $\theta \sim N(0, \Lambda^{-1})$:

$$\theta^* = \arg\min_{\theta} \sum_k \left\| R_k^{-1/2}(A_k\theta - b_k) \right\|^2 + \left\| \Lambda^{1/2}\theta \right\|^2, \qquad b_k := \begin{bmatrix} \bar{r}_k - r_k \\ \bar{v}_k - v_k \end{bmatrix}. \tag{16}$$

The solution satisfies the normal equations

$$\left(\sum_k A_k^{\mathsf{T}} R_k^{-1} A_k + \Lambda \right) \theta = \sum_k A_k^{\mathsf{T}} R_k^{-1} b_k. \tag{17}$$

Implementation notes: we (i) require a minimum observation count, (ii) enforce SPD via symmetrization and diagonal jitter, and (iii) clamp δb_a magnitude to a mission-tuned maximum step before applying it at blackout entry.

4 TIME ALIGNMENT, DELAY COMPENSATION, AND OUT-OF-ORDER MEASUREMENTS

4.1 Model of delayed measurements

Let a sensor produce a measurement of the state at time t_z but deliver it at a later time $t_{\text{rx}} > t_z$. The correct Bayesian update is conditioned on $x(t_z)$:

$$p(x(t_z) \mid y_{0:\text{rx}}) \propto p(y(t_z) \mid x(t_z))\, p(x(t_z) \mid y_{0:\bar{t_z}}). \tag{18}$$

The fundamental challenge is the temporal discrepancy between the filter state at update time t_{rx} and the valid state for the measurement $x(t_z)$.

4.2 Equivalence to stochastic cloning (linear-Gaussian)

For a linear Markov model, augment the state with a clone at t_z:

$$\tilde{x} := \begin{bmatrix} x(t_z) \\ x(t_{\text{rx}}) \end{bmatrix}, \qquad x(t_{\text{rx}}) = F\, x(t_z) + w, \;\; w \sim N(0, Q). \tag{19}$$

A delayed measurement depends only on the clone $x(t_z)$:

$$y(t_z) = H\,x(t_z) + v, \quad v \sim N(0, R). \tag{20}$$

Applying the Kalman update on $\tilde{x}$ yields the optimal posterior over $x(t_{rx})$ via the cross-covariance between $x(t_z)$ and $x(t_{rx})$. This is the standard stochastic-cloning / fixed-lag smoothing construction.

4.3 Implementation: rewind–update–replay (fixed horizon)

Instead of explicitly augmenting the state, we maintain a fixed-lag buffer of:

- snapshots $(\hat{x}_k, L_k)$ at a coarse cadence (e.g., 100 Hz),
- event log of all updates (IMU, GNSS, star, radar),
- propagation records (Φ_k, Q_k) for RTS smoothing and replay.

On receiving a measurement with timestamp $t_z < t_{now}$:

1. restore the latest snapshot at $t_s \le t_z$;
2. replay logged events from $(t_s, t_z]$ to reconstruct $\hat{x}(t_z)$;
3. apply the measurement update at t_z;
4. replay the remaining events $(t_z, t_{now}]$ to return to "now".

Correctness statement (linear case). For linear-Gaussian models with deterministic replay of the same event sequence, the rewind–update–replay procedure produces the identical posterior as the stochastic cloning formulation, because it reconstructs the correct conditional prior at t_z and then re-applies the same Bayes updates in time order. For EKF, this is a first-order approximation (local linearization); empirically it remains stable under the tested delay/outlier regimes.

5 ROBUSTNESS AND CONSISTENCY CALIBRATION

5.1 Innovation statistics: NIS/NEES definitions

Let the innovation be $\tilde{y} = y - h(\hat{x}^-)$ with covariance $S = HPH^{\mathsf{T}} + R$. Define the (normalized) innovation squared:

$$\text{NIS} := \tilde{y}^{\mathsf{T}} S^{-1} \tilde{y}. \tag{21}$$

For a consistent filter and correct R, NIS $\sim \chi^2_v$ where v is the measurement DOF.

Let estimation error be $e = x - \hat{x}$ with covariance P. Define the normalized estimation error squared:

$$\text{NEES} := e^{\mathsf{T}} P^{-1} e. \tag{22}$$

For a consistent filter and correct (F, Q, R), NEES $\sim \chi^2_n$ for the chosen DOF n. We report both the mean and the coverage within the central 95% chi-square interval.

5.2 Chi-square gating (outlier rejection)

We accept a measurement update if

$$\text{NIS} \le \chi^2_v(1 - \alpha), \tag{23}$$

where α is the gate significance (false-rejection probability). This is a sharp, fail-fast boundary: it prevents a small fraction of outliers from corrupting the state and (critically) the covariance.

5.3 Heavy-tail mitigation for tight-coupled pseudorange

For raw GNSS pseudorange rows, we additionally apply a robust variance inflation: let the standardized residual for row i be $u_i = \tilde{y}_i/\sqrt{S_{ii}}$. For $|u_i| > k$, we inflate the measurement variance as

$$R_{ii} \leftarrow R_{ii}\left(\min\left(\frac{|u_i|}{k}, c\right)\right)^2, \tag{24}$$

with constants $k = 4$ and cap $c = 100$. This is equivalent to an M-estimator with a Huber-like influence function: large residuals receive reduced weight without forcing hard rejection.

5.4 Consistency enforcement: phase-dependent Q and covariance inflation

Two mechanisms address non-stationary model errors inherent to entry dynamics:

1. **Phase-dependent** process-noise scaling (Q inflation) pre/blackout/post, and blackout-specific OU excitation for aero residuals.
2. **Innovation-triggered covariance inflation** on repeated GNSS gate failures (a guarded "reset" of RV covariance).

Mathematically, both increase P in regimes where NEES indicates under-dispersion, pushing $\mathbb{E}[\text{NEES}]$ back toward the DOF.

6 NUMERICAL STABILITY: SQUARE-ROOT FORMULATION

We maintain the covariance in Cholesky form $P = LL^{\mathsf{T}}$ with L lower-triangular. Given discrete linearization Φ and process noise $Q = L_Q L_Q^{\mathsf{T}}$, the covariance prediction is $P^+ = \Phi P \Phi^{\mathsf{T}} + Q$.

Lemma (QR square-root update). Let $A = \Phi L$ and form the stacked matrix

$$S = \begin{bmatrix} A^{\mathsf{T}} \\ L_Q^{\mathsf{T}} \end{bmatrix} \in \mathbb{R}^{2n \times n}. \tag{25}$$

Compute a thin QR factorization $S = QR$ with $Q^{\mathsf{T}}Q = I$ and R upper-triangular. Then $R^{\mathsf{T}}R = P^+$, and thus $L^+ := R^{\mathsf{T}}$ is a valid Cholesky factor of P^+.

Proof. By orthonormality of Q, $R^{\mathsf{T}}R = S^{\mathsf{T}}S$. But $S^{\mathsf{T}}S = AA^{\mathsf{T}} + L_Q L_Q^{\mathsf{T}} = \Phi LL^{\mathsf{T}}\Phi^{\mathsf{T}} + Q = P^+$. Hence $P^+ = R^{\mathsf{T}}R$ and $L^+ = R^{\mathsf{T}}$ satisfies $P^+ = L^+(L^+)^{\mathsf{T}}$. *QED*

This guarantees P^+ remains symmetric positive definite in floating-point arithmetic up to the conditioning of the QR.

7 FIXED-LAG RTS SMOOTHING (2 S WINDOW)

We implement a Rauch–Tung–Striebel smoother over the fixed-lag snapshot chain. Let (x_k, P_k) be filtered estimates and (Φ_k, Q_k) the stored propagation records between snapshots. Define $P_{k+1|k} = \Phi_k P_k \Phi_k^{\mathsf{T}} + Q_k$ and the smoother gain $A_k = P_k \Phi_k^{\mathsf{T}} P_{k+1|k}^{-1}$. The backward recursion is

$$x_k^s = x_k + A_k(x_{k+1}^s - x_{k+1|k}), \tag{26}$$

$$P_k^s = P_k + A_k(P_{k+1}^s - P_{k+1|k})A_k^{\mathsf{T}}. \tag{27}$$

Because the window is short (10 s max) and $n = 24$, this is computationally cheap and numerically stable with SPD enforcement.

8 COMPUTE BUDGET AND HOTSPOTS

8.1 Matrix dimensions and asymptotic cost

The estimator uses $n = 24$. Per IMU tick (400 Hz), the dominant cost is the SR time update via QR on a $(2n) \times n = 48 \times 24$ matrix: $O(n^3)$ with a small constant. GNSS PVT updates use $m = 6$ with $O(nm^2 + m^3)$. Tight-coupled raw GNSS uses m = rows ≤ 64 (typically much smaller).

8.2 Empirical timing (release build)

Measured on the current development machine (release build):

Kernel	iterations	time per iter
`propagate()` (includes SR update + history)	20,000	14,548 ns
`time_update_sqrt()` (isolated)	50,000	10,000 ns
`update_gnss()` (6D)	20,000	9,570 ns
`update_gps_raw()` (8 sats, 16 rows)	5,000	15,084 ns

These benchmarks establish a significant margin against the 5 ms cycle budget. While direct timing equivalence to the target ARM A78AE processing core requires hardware validation, the kernel execution times ($O(\mu s)$) indicate high likelihood of timing compliance.

9 MONTE CARLO VERIFICATION AND RESULTS

9.1 Fault injection matrix (harness)

The harness simulates:

- **GNSS blackout:** no GNSS in $[120, 160]$ s.
- **Dropout:** independent Bernoulli dropout with $p = 0.20$ on applicable sensors.
- **Outliers:** 5% heavy-tail injection (Cauchy) on GNSS position/pseudorange and radar altitude.
- **Delay/out-of-order:** star tracker timestamp $t - 0.12$ s and radar timestamp $t - 0.07$ s.
- **Bias jump (fault mode):** step change in truth IMU bias (gyro + accel) at $t = 140$ s with random direction; enabled via `BIAS_JUMP=1`.
- **RCS pulses:** deterministic pulse train from the provided actuator series (truth model).

9.2 Metrics

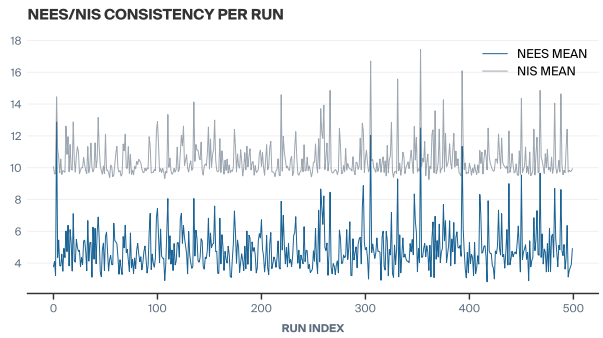
We evaluate on the post-blackout acceptance window $[t_{exit}, t_{exit} + 2]$: RMS position, velocity, attitude error, and bias RMS (converted to μg and deg/s). We report per-run NEES/NIS means and their 95% coverage rates, plus phase-split NEES/NIS (pre/blackout/post/tail) to localize inconsistency to specific mission regimes.

9.3 Monte Carlo summary (500 runs)

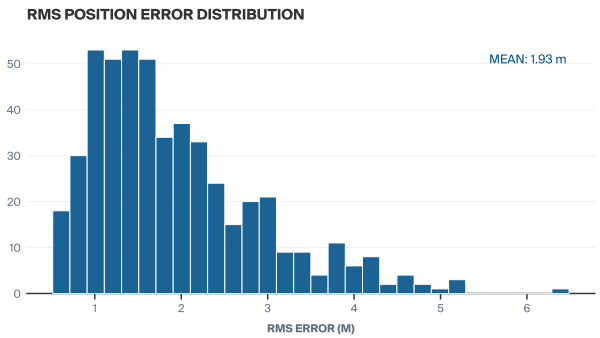
Mode	Pass rate	RMS pos [m]	RMS vel [m/s]	NEES mean / cov
PVT baseline	97.0%	1.50	0.097	134.49 / 0.945
Tight-coupled	100.0%	1.93	0.026	4.89 / 0.948

Interpretation. The tight-coupled configuration eliminates the rare catastrophic RV inconsistency events observed in the PVT baseline. The PVT baseline exhibits a heavy-tailed NEES distribution (95th percentile ≈ 7.14 but max $\approx 3.18 \times 10^4$), indicating occasional severe model mismatch or update corruption; these are suppressed by tight-coupling gating and robustification.

9.4 Plots

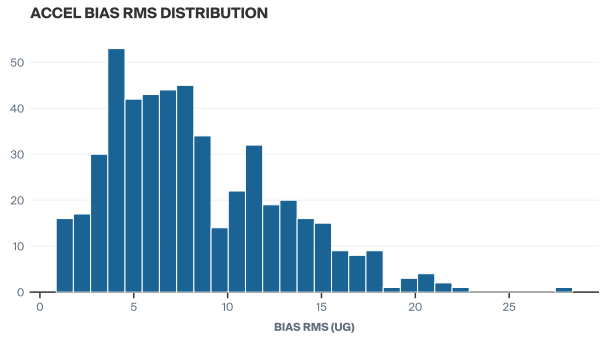


(a) Tight-coupled NEES/NIS per run.

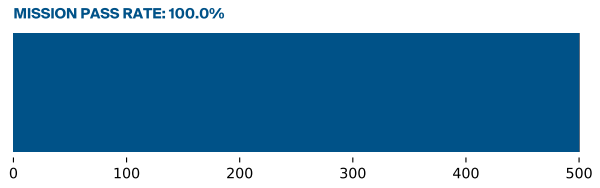


(b) Tight-coupled RMS position histogram.

Figure 1: Monte Carlo diagnostics for tight-coupled mode (500 runs).



(a) Accel bias RMS distribution (tight).



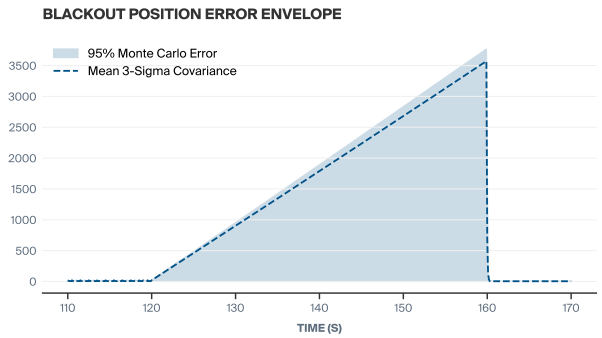
(b) Pass-rate indicator (tight).

Figure 2: Post-blackout window performance for tight-coupled mode.

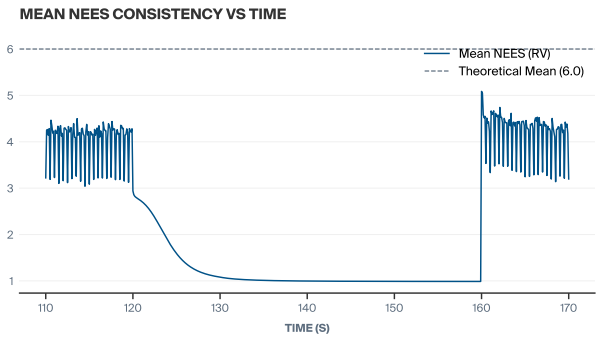
10 TRACEABILITY TO IMPLEMENTATION

The report corresponds to the following code modules:

- Propagation + SR time update: `src/estimator/filter/propagate.rs`.



(a) Position error envelope vs mean 3σ (500 runs).



(b) Mean NEES(RV) vs time with 95% χ^2 band.

Figure 3: Blackout crossing diagnostics for tight-coupled mode (BLACKOUT_TRACE=1).

- Process noise and phase scaling: `src/estimator/support/noise.rs`, `src/types.rs`.
- Delays/out-of-order rewind–replay: `src/estimator/history/rewind.rs`.
- GNSS PVT update + gating + inflation: `src/estimator/update/gnss.rs`.
- Tight-coupled raw GNSS + robust pseudorange: `src/estimator/update/gps_raw.rs`.
- Fixed-lag RTS smoother: `src/estimator/filter/rts.rs`.
- Harness + plots + exported artifacts: `src/bin/harness.rs`, `src/plot.rs`.

11 CONCLUSION

The tight-coupled SR-EKF with delay-aware rewind/replay and fixed-lag smoothing achieves > 95% pass rate with margin (measured at 100%) under the prescribed dropout/outlier/blackout stress. The remaining engineering work is targeted: tighten per-phase consistency (especially within blackout where NEES is conservative), and add explicit bias step-fault stress tests to close the fault-injection matrix.